

AN EVALUATION OF STUDENT ENGAGEMENT AND EXPERIENTIAL LEARNING USING ENTERPRISE APPLICATION TESTING

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Abstract

Experiential learning provides a mechanism for producing work ready graduates. Undergraduate curriculum, which provide opportunities for professional practice are especially suitable for resource constrained economies. However, one challenge is how best to blend the curriculum of practice with the core curriculum in a way that provides meaningful experience and active learning without diminishing the core theoretical frame of the program.

This study pilots an experiential learning approach in an undergraduate Software Implementation and Testing course. The pilot involved students testing and actively interacting with an Enterprise application. The purpose of the pilot was to provide real-world experience in order to enhance students' software testing capabilities and knowledge.

Using a qualitative approach, the study evaluates students' knowledge about specific aspects of software testing before and after the pilot. The feedback provided was compared and evaluated to determine what students learned from the experience and how their skills were enhanced.

This strategy was found to improve active learning, and in addition to sharpening students' software testing skills. However, there were challenges in terms of the learning curve for various aspect of the testing tool. Feedback from the pilot can be used to improve the experiential component of the course.

Keywords: Experiential Learning, Active Learning, Software Testing.

1 INTRODUCTION

The Digital firms are dependent on reliable, high quality and highly functional software to support and sustain their enterprises [1]. Failure of software supporting these enterprises can be disastrous and can result in loss of revenue, paralysis of critical operations and ultimately failure of the enterprise. Software testing is one of the primary approaches that developers use to produce reliable software and is a critical component of the software quality assurance curriculum in undergraduate computing curriculum. Unfortunately, often time insufficient emphasis is placed on software testing in the computing curriculum of universities [2]. According to [3] due to the nature of some undergraduate programs students do not get the opportunity to engage in real world projects that can provide exposure to principles of software engineering. In addition companies are not willing to sacrifice their projects by having students working on them [3]. So [3] advocates for the use of free and open software applications and projects as a mechanism for giving students the life long learning exposure to real world software engineering principles that will improve their education. In this project we provide another method that can be used by educators to help students to develop their competencies in software testing.

In fact [4] blames the shortage of specialized professionals in the software testing industry on the inefficiency in software testing education, as characterized by the absence of structured testing environments and methods. The authors posits that students do not get an integrated view of software testing since in the undergraduate programs, different components of software testing are presented in different courses [4]. In addition students do not acquire the attitude for rigorous testing, which is needed for software product development. Furthermore, students do not see the relationship between testing and courses such as Algorithms, Data Structures, Software Requirements Specification etc. [4]. In addition, software testing is done only at the end of the undergraduate program of study and does not give sufficient time for students to leave the program ready for testing in industry [4].

The value of experiential learning is well documented. Accordingly [5] defines experiential learning as a philosophy and methodology that engages learners in direct experience and purposed reflection in order to deepen knowledge and develop skill. While [6] proffers that experiential learning activities provide concrete, real-world types of experiences to computing majors, extending their learning opportunities and strengthening their educational experience. It provides a bridge that allows students to mature in their understanding of customer needs [6]. Experiential learning helps to increase students' confidence, since they see how their work is put into use [6]. In addition [6] also points out that according to experiential learning theory, concrete experiences form the basis for observations and reflection upon which new understandings can emerge. Interestingly [3] completed a pedagogical investigation using an open source framework for software testing, with notable success. Through an experiential learning strategy, the framework enabled students to test code that they did not write. They found that the framework heightened the interest of students in software testing, and brought industry experience to the student. They also found that it is possible to integrate real world experiences into the formal teaching and learning structure.

In this study students were introduced to a web-based enterprise application system that was at the beta-testing phase of development. The application was earmarked for implementation across the island, and was deemed suitable for the pilot. The aim of the pilot was to give students some insight and experience into the role of software testers while they were enrolled in a course in software implementation and testing. Like other universities, elements of software testing and quality assurance are dispersed among the various programming courses in the curriculum. Feedback from past students and prospective employers suggested that students were ill prepared to handle software testing responsibilities. The pilot project was carried out in an effort to (1) assess the students perceived readiness for software testing and (2) to provide students with an opportunity to gain experience as software testers of a real world enterprise type software application,

2 METHODOLOGY

This research project used an exploratory pilot study to evaluate the students' learning and impression after using an enterprise application for software testing. Qualitative data was collected. At the start of the study students were given a preliminary questionnaire that sought to capture their knowledge and impression of software testing. The students then did the Software Implementation and Testing course. As part of the course the students were required to test the enterprise application. Before testing begun, students attended an information session and demonstration of the enterprise application. During this session, students were given details on the application's purpose, features and background. There was also a question and answer period. Details of the system's functionalities in terms of user groups were elaborated. The primary purpose of this session was to introduce the students to the system. In addition after the information session documentation on the system was sent to the students.

A separate test bed was created by the system administrator for the testing environment in order to ensure that a consistent version of the application was provided. Students were required to develop test cases and to devise their own testing strategy. They recorded their developed test cases using a template provided by the instructor. Students were required to log into the enterprise application via the internet, test and record their test results. Both their developed test cases and their recorded test results were graded separately. At the end of the course students were given the same questionnaire which was administered at the start of the study. In addition the students were asked additional questions in order to capture their impression of the testing experience. The data gathered was analyzed and recorded.

3 RESULTS

During week six of the semester, before the experiential learning activity begun, students were asked to complete a short questionnaire. The class was comprised of about fifteen students, nine willing participated in the pre-survey while eleven willing participated in the post survey and feedback on the experiential learning activity.

3.1 Pre-Survey

This section details the findings from the initial survey. When asked how would you approach testing the application to ensure that all bugs are removed, the initial response of some students suggested that they had some knowledge of different types of testing but lacked specific knowledge, skill and structured process. Their responses included:

"I would use monkey testing"

"I would use a stress test method to ensure that the system can handle a large amount of data input"

However, some students seem to have had a better understanding:-

"As developers we can break down the application into simple tasks and test each task..."

"I would perform functional testing then develop test cases for the system"

"I would split it into sections and test them. Also I would integrate the sections that work together and test them with each other for errors"

"...create a test case for each function based on user documentation...final step would be to create automated test."

However, the most varied and unstructured response was:

"By conducting an alpha testing to remove some of the defects...by ensuring that specification and requirement functions are doing what they should do...as well as beta testing...before doing these testing the white box testing should be done... and black box testing done by users."

Just over half of the group felt inadequately prepared to lead a team to carry out this [testing] exercise. Those who did not feel confident realized that they would have to know more about the application requirements and testing process. One student felt that he lacked the technical ability but indicated that he was willing to learn. Another felt that he was not innately a leader. When asked how well prepared do you feel about taking responsibility for testing a segment of the application, most students felt somewhat prepared. One student expressed that he had some experience in testing. Another stated that he would have to do a test plan. Others were less confident as evidenced by:-

"If I fully understand how the model should work I would be prepared"

"Not well but willing to research and educate [myself] oneself"

Students were asked which course or area in the curriculum best prepared them to undertake the current software testing task. The majority of the students felt that the current Software Implementation and Testing course best prepared them. Two students felt that in addition to the Software Implementation and Testing course all the programming courses prepared them for the task. One student felt that the Web Programming course best prepared him for the task at hand. Students were also asked where in the curriculum they think would be the best place for students to develop their software testing skills. Some of the responses seem to indicate some confusion as to what exactly was being asked. Three students felt that the best place was in the current course, one student indicated that it should be in the Object oriented course, yet another student thought that there should be lab sessions and two students felt that it should be taught as early as possible in the program:

"I believe it should be introduced from the very beginning, being able to perform successful testing would aid in all the programming courses,...I personally recommend introducing it after Programming 2 [at the end of year 1]."

The last questions on the preliminary survey asked students if they have considered Software Quality Assurance as a possible profession that they would pursue and why or why not. Three student said "no" that they had not considered it, two students said "yes" one student said possibly and others left the question blank. Reasons given why they would or would not considered are shown below:

Table 1 – Would you consider Software Quality Assurance as a Profession?

Answers	No	Yes	Possible
Reasons Given ↓ ↑	"I want to be a business analyst"	"Yes I have before at the time it seemed like an easy task to do."	"It is a good profession improving performance and reliability in a software and have good feedback from users."
	"I am best at backend coding"	"The area seem[s] somewhat exciting being able to navigate code whether via script or manually..."	
	Too much work to ensure the software is functioning properly.		

3.2 Post Survey

After the experiential activity was completed at the end of the semester the students were given the same questions again, with the aim to determine how much they valued the experiential learning activities. This section details the results from the second survey.

When asked how would you now approach testing future applications to ensure that all bugs are removed, students spoke about the need to understand the functional and nonfunctional requirements of the application before starting the testing; the importance of testing all branches of the application; the need to develop good test cases; the need to use a mix of testing strategies (manual and automated) and the importance of the testing tool.

When asked how well prepared do you feel about leading a team in conducting this exercise, all felt prepared except for one student who felt "not very prepared". Those who felt prepared to lead had feedback such as:

"I feel prepared enough to take on the challenge of spearheading a team to test an application as there is a new appreciation to the dynamics of testing an application."

"I would be prepared to lead a team in doing this type of test, because I had hands on experience in doing such and learning how to do this type of test I'm more familiar with this test method."

To the question how well prepared do you feel about taking responsibility for testing segments of an application, again all but one student felt prepared with responses such as:-

"With the methods and skills I've learned throughout the course I'm confident in my ability to test and take responsibility of a complete segment of a system."

I think I would be well prepared. I learnt the approaches and I got the hands on experience and I also got the theory of testing."

When asked which modules or area of the program best equipped you for undertaking the testing task most students felt that the current course Software Implementation and Testing along with the Programming courses best prepared them. One student mentioned the database course along with the two already mentioned. When asked where in the program would be the best place for software testing skills to be developed in the students, two students thought it should be when students are learning to code, one student said "1st year", and a couple students said the software implementation and testing course. To the question have you considered Software Quality assurance as a possible profession you could pursue about half of the students said "yes", this was more than indicated before the experiential learning activity.

3.3 Feedback on the Experiential Learning Activity

Additional questions were asked to find out how students felt about the experiential learning activity. To the questions of how useful was the enterprise system testing activity all students felt that it was useful, as they were able to put what they were learning into practice. Feedback included

"It was very helpful and very informative. As a teaching tool the practical and hands on aspect of the exercise allowed for a better appreciation of testing..."

The question of what did you learn or what new knowledge did you gain from the testing activity yielded mention of some tangible abilities and behavioral attributes they acquired for example:

"I learnt how to identify bugs in applications and to ensure to locate bugs or errors at the first stage"

"In addition to learning about testing techniques I learnt about automated testing software. I also see [better appreciated] that testing is a very important in software development."

Several students stated that they learn how to construct test cases. One student felt that the exercise gave him a better understanding of what he would be doing in the specialization he intended to do. One student felt that there was no new information just a different appreciation:

"No explicitly new information was gained, however a different appreciation on previously gained knowledge was achieved in seeing upfront a system in development that further testing could reveal minor issues with major implications on the system as a deliverable."

All students indicated that they would recommend this "real world" approach for upcoming classes. The reasons given for this recommendation included:-

"It's a more hands on [approach], it's easier for students to learn from and less complex than the white box testing."

"It's a good way to gain experience"

"A practical appreciation for testing is always readily more applicable than a predominantly theoretic approach. After all the students are being prepared for a practical working world."

To the question of what was good about the approach, students felt that the approach was good because it required them to find suitable strategy for testing the system, write test cases and report on their findings. It also enabled them to see firsthand the result of testing and the value of a system being test by someone other than the developer.

It appeared that many principles and common sense practices became apparent to the students based on their response to the question of what are the lessons or take away points from the experience, evidenced by responses such as:

"Have a general knowledge about the system you are testing"

"Start early and do a little everyday because it can get overwhelming"

"Apart from programming this course allowed me to feel like I'm gaining work experience"

"It's easier to identify bugs when we use software tools instead of using strictly coding"

"Validation is key to app development"

"Test cases are important so you know what you are looking for and the developer would need the feedback to help proper functionality of the application"

"Record all findings large and small"

Finally students were asked to suggest any other strategies or approaches that can be employed to help students' learning. Some of their responses were:-

"Maybe it can be implemented in programming projects as an option...Students should be allowed to test a separate system from the one they build..."

"Teach students to use the software tools to help in testing"

"Apply this approach to other course areas..."

"Clouding the contents of testing into a Software Implementation course has time constraints that may make it difficult to be grasped fully. An independent practical course on the area would be better suited to study in finer detail newer and varying testing technologies."

3.4 Discussion

The findings support the literature, which indicated that testing concepts should not be left to one course towards the end of the program, but rather should be introduced at the start of the program. Although students are taught some practical elements of testing in different programming courses, when confronted with a real world enterprise system they generally lacked confidence, cohesive strategy and methodology to attack the problem. Experiential learning helps to address this challenge, as demonstrated, the experience provided the students with valuable insight and the opportunity to reflect. This too is in keeping with the literature. In addition there are some benefits to having students interact with enterprise type systems similar to those they will meet in industry rather than just small program that they develop. Surprisingly although the exposure time to the enterprise system spanned over a few week the activity was still quite meaningful.

4 CONCLUSIONS

Due to the small sample size it is not possible to make sweeping generalizations, nevertheless some useful insights were garnered. It is clear that there is a need to better coordinate the programming and software engineering courses to ensure that from early in the program students learn core concepts and systematic practices for testing. There should also be a greater consideration of the tools and practices used for testing. In addition adding an experiential enterprise type activity not only provides students with hands on real world experience but it also provides them with behavioral insights.

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